



10968 - Where are the clouds? Mapping the molecular gas content in the extraordinary steady state feedback galaxy group NGC 5813

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Scott Randall (PI)	Smithsonian Institution Astrophysical Observatory
Dr. Grant R. Tremblay (CoI) (CoPI)	Smithsonian Institution Astrophysical Observatory
Dr. Ewan O'Sullivan (CoI)	Smithsonian Institution Astrophysical Observatory
Dr. Elizabeth Blanton (CoI)	Boston University
Dr. Gerrit Schellenberger (CoI)	Smithsonian Institution Astrophysical Observatory
Dr. William R. Forman (CoI)	Smithsonian Institution Astrophysical Observatory

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Core B Band	MIRI Medium Resolution Spectroscopy	(1) NGC-5813
	3	Core A Band	MIRI Medium Resolution Spectroscopy	(1) NGC-5813
	5	Core C Band	MIRI Medium Resolution Spectroscopy	(1) NGC-5813
	2	Local Blank Sky Backg round B Band	MIRI Medium Resolution Spectroscopy	(2) NGC_5813_BG
	4	Local Blank Sky Backg round A Band	MIRI Medium Resolution Spectroscopy	(2) NGC_5813_BG
	6	Local Blank Sky Backg round C Band	MIRI Medium Resolution Spectroscopy	(2) NGC_5813_BG

ABSTRACT

We propose JWST/MIRI Medium Resolution Spectrometer (MRS) observations of the central region of the nearby group-dominant elliptical galaxy NGC 5813 to map the warm molecular hydrogen (H) associated with AGN-driven feedback. NGC 5813 exhibits a unique system of X-ray cavities and shock fronts, indicating that it is in a "steady state" feedback mode with a stable and known kinetic AGN feedback power for at least the past 50 Myr. Extended H-alpha filaments trace uplifted cavities and cool X-ray gas, as well as [CII] emission, indicating a connection from the hot X-ray and ionized phases down to low-density atomic or molecular gas. However, the molecular component connecting these phases remains poorly constrained. Surprisingly, CO emission, the most commonly observed tracer of the cold molecular component predicted by the chaotic cold accretion model, remains undetected despite deep observations, underscoring the need for direct H₂ observations. Using three MRS sub-band exposures, we will obtain spatially resolved spectroscopy of the H 0–0 S(6)–S(1) lines, providing a direct census of the warm (T 200–2000 K) molecular gas phase. The observations will map line fluxes, excitation temperatures, and kinematics across the and core, determining the molecular gas mass and testing whether its distribution and properties are consistent with the chaotic cold accretion model. Only JWST is capable of making these measurements in this spectacular steady state feedback galaxy group.

OBSERVING DESCRIPTION

The program will obtain MIRI MRS observations of the central 3×3 region of NGC 5813 to detect and map H 0–0 S(6)–S(1) emission across the full thermal ladder. We will use three sub-band settings (1A–3C) to cover key diagnostic lines H S(6) 6.11 μm , S(5) 6.91 μm , S(4) 8.03 μm , S(3) 9.66 μm , and S(1) 17.03 μm . Each setting will employ the standard four-point dither pattern to ensure Nyquist sampling and mitigate bad pixels. Exposure times of 3300 s per band (55 min total, including 50 % offset background) yield expected signal-to-noise ratios of 15–30 per beam, sufficient to measure line fluxes, ratios, and velocity widths ($\sim 10 \text{ km s}^{-1}$) for all detected lines.

Because NGC 5813 is an extended source with filamentary emission, we will obtain dedicated offset sky backgrounds located several arcmins away from the galaxy, outside the X-ray halo and optical extent. This strategy minimizes contamination from faint diffuse emission while allowing accurate subtraction of the strong thermal background at $> 10 \mu\text{m}$. Each background exposure will use the same dither and readout parameters as the science exposure. Background and source observations are grouped as non-interruptable by band, to mitigate the impact of short, $\sim$ hour long variations in the background. Thus, we propose 3 visits, one for each A/B/C band, with the background and source observations performed in the same uninterruptable sequence.

For target acquisition, we will use the MIRI Imager with the F1130W filter, which provides optimal throughput and contrast for early-type galaxies. We adopt the FAST readout pattern with 6 groups/integration and 2 integrations, giving a reliable centroid without saturating the nucleus. Simultaneous Imager data will improve astrometric registration between the MRS cube and optical/IR maps.

Proposal 10968 - Targets - Where are the clouds? Mapping the molecular gas content in the extraordinary steady state feedback gala...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC-5813	RA: 15 01 11.2720 (225.2969667d) Dec: +01 42 6.45 (1.70179d) Equinox: J2000	Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Active galactic nuclei, Brightest cluster galaxies, Elliptical galaxies, Galaxy nuclei] Extended=YES				
	(2)	NGC_5813_BG	RA: 15 01 18.8400 (225.3285000d) Dec: +01 44 39.50 (1.74431d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background]				

Proposal 10968 - Observation 1 - Where are the clouds? Mapping the molecular gas content in the extraordinary steady state feedbac...

Observation	Proposal 10968, Observation 1: Core B Band												Fri Mar 13 20:04:24 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[Local Blank Sky Background B Band (Obs 2)]												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	NGC-5813		RA: 15 01 11.2720 (225.2969667d) Dec: +01 42 6.45 (1.70179d) Equinox: J2000			Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Galaxy												
	Description=[Active galactic nuclei, Brightest cluster galaxies, Elliptical galaxies, Galaxy nuclei] Extended=YES												
Acquisition	#	Target		Filter	Readout Pattern		Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID	
	1	1 NGC-5813		F1000W	FAST		10	1		1	27.75	271344	
Template	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	All MRS			YES			FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1130W	FASTR1	100	2	1	Dither 1	4	8	2231.132	
	1	MEDIUM(B)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
	1	MEDIUM(B)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	

Special Requirements	Group Observations 1, 2, Non-interruptible
----------------------	--

Proposal 10968 - Observation 3 - Where are the clouds? Mapping the molecular gas content in the extraordinary steady state feedbac...

Observation	Proposal 10968, Observation 3: Core A Band												Fri Mar 13 20:04:24 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[Local Blank Sky Background A Band (Obs 4)]													
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous				
	(1)	NGC-5813		RA: 15 01 11.2720 (225.2969667d) Dec: +01 42 6.45 (1.70179d) Equinox: J2000			Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Category=Galaxy													
	Description=[Active galactic nuclei, Brightest cluster galaxies, Elliptical galaxies, Galaxy nuclei] Extended=YES													
Acquisition	#	Target		Filter	Readout Pattern		Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time		Optional ETC ID	
	1	1 NGC-5813		F1000W	FAST		10	1		1	27.75		271344	
Template	Primary Channel			Simultaneous Imaging				Imager Subarray			Grating Wheel Direction			
	All MRS			YES				FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F1130W	FASTR1	100	2	1	Dither 1	4	8	2231.132		
	1	SHORT(A)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132		
	1	SHORT(A)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132		

Special Requirements	Group Observations 3, 4, Non-interruptible
----------------------	--

Proposal 10968 - Observation 5 - Where are the clouds? Mapping the molecular gas content in the extraordinary steady state feedbac...

Observation	Proposal 10968, Observation 5: Core C Band												Fri Mar 13 20:04:24 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[Local Blank Sky Background C Band (Obs 6)]												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	NGC-5813	RA: 15 01 11.2720 (225.2969667d) Dec: +01 42 6.45 (1.70179d) Equinox: J2000				Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Galaxy												
	Description=[Active galactic nuclei, Brightest cluster galaxies, Elliptical galaxies, Galaxy nuclei] Extended=YES												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID			
	1	1 NGC-5813	F1000W	FAST	10	1		1	27.75	271344			
Template	Primary Channel		Simultaneous Imaging				Imager Subarray			Grating Wheel Direction			
	All MRS		YES				FULL			Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1130W	FASTR1	100	2	1	Dither 1	4	8	2231.132	
	1	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
	1	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	

Special Requirements	Group Observations 5, 6, Non-interruptible
----------------------	--

Proposal 10968 - Observation 2 - Where are the clouds? Mapping the molecular gas content in the extraordinary steady state feedbac...

Observation	Proposal 10968, Observation 2: Local Blank Sky Background B Band												Fri Mar 13 20:04:24 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [Core B Band (Obs 1)]												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(2)	NGC_5813_BG	RA: 15 01 18.8400 (225.3285000d) Dec: +01 44 39.50 (1.74431d) Equinox: J2000										
	Comments: Category=Calibration Description=[Telescope/sky background]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1000W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					BACKGROUND			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1130W	FASTR1	100	1	1	Dither 1	4	4	1110.016	
	1	MEDIUM(B)	MRSLONG		FASTR1	100	1	1	Dither 1	4	4	1110.016	
	1	MEDIUM(B)	MRSSHORT		FASTR1	100	1	1	Dither 1	4	4	1110.016	

Special Requirements	Group Observations 1, 2, Non-interruptible
----------------------	--

Proposal 10968 - Observation 4 - Where are the clouds? Mapping the molecular gas content in the extraordinary steady state feedbac...

Observation	Proposal 10968, Observation 4: Local Blank Sky Background A Band												Fri Mar 13 20:04:24 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [Core A Band (Obs 3)]												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	NGC_5813_BG	RA: 15 01 18.8400 (225.3285000d) Dec: +01 44 39.50 (1.74431d) Equinox: J2000										
	Comments: Category=Calibration Description=[Telescope/sky background]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
	F1000W	All MRS			YES			FULL			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1130W	FASTR1	100	1	1	Dither 1	4	4	1110.016	
	1	SHORT(A)	MRSLONG		FASTR1	100	1	1	Dither 1	4	4	1110.016	
	1	SHORT(A)	MRSSHORT		FASTR1	100	1	1	Dither 1	4	4	1110.016	

Special Requirements	Group Observations 3, 4, Non-interruptible
----------------------	--

Proposal 10968 - Observation 6 - Where are the clouds? Mapping the molecular gas content in the extraordinary steady state feedbac...

Observation	Proposal 10968, Observation 6: Local Blank Sky Background C Band												Fri Mar 13 20:04:24 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [Core C Band (Obs 5)]												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	NGC_5813_BG		RA: 15 01 18.8400 (225.3285000d) Dec: +01 44 39.50 (1.74431d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
	F1000W	All MRS			YES			FULL		Allow Auto Reorder			
Dithers	#	Dither Type			Optimized For			Direction					
	1	4-Point			POINT SOURCE			NEGATIVE					
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1130W	FASTR1	100	1	1	Dither 1	4	4	1110.016	
	1	LONG(C)	MRSLONG		FASTR1	100	1	1	Dither 1	4	4	1110.016	
	1	LONG(C)	MRSSHORT		FASTR1	100	1	1	Dither 1	4	4	1110.016	

Special Requirements	Group Observations 5, 6, Non-interruptible
----------------------	--